

Demand Transmission Between Sectors

Question: Do increases in new orders for fabricated metals predict shipments of primary metals?

- **H₀:** New orders in fabricated metals do not predict shipments in primary metals.
- **H₁:** There is a significant positive lagged relationship.

Test:

- **OLS regression** on lagged orders

ML extension:

Train an XGBoost regression model to predict shipments using lagged orders; test feature importances to confirm significance.

Construction Demand → Metal Production

Question: Does growth in residential construction (resconst) predict shipments in the primary metals M3 sector?

- **H₀:** Construction permits/starts have no effect on metal shipments.
- **H₁:** Construction activity leads to higher metal shipments within 1–3 months.

Tests:

- **Dynamic regression / distributed lag model**

ML extension:

Use lagged construction activity and macro factors (CPI, industrial production) to forecast metal shipments; measure predictive lift.